

Unit 4 :: Data Preparation and Management

☞ SEC-03-PHY-91-R-03 ☞

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Syllabus

Unit 4: Data preparation and management

Basic pre-processing of data: filtering, sorting, normalization and handling missing values; managing multisheet workbooks and structured datasets, importing data from diverse file formats (CSV, ASCII, DAT), and ensuring interoperability between different spreadsheet and analysis platforms.

1 Importing data from diverse file formats

Suppose a physics experiment records a variable which is of interest to us, such as temperature, pressure, voltage, count rate etc. These variables are recorded (or sampled) at a sequence of time instants. Now this raw record of data produced by an instrument is almost certainly unfit for direct analysis because

- a) It may contain unreliable readings.
- b) It may be out of chronological order.
- c) It may mix standard and non-standard units.
- d) It may contain gaps.

Then we say that such a dataset must be *pre-processed* before it can be visualised or fitted to a theoretical curve.

For the purpose of pre-processing, the dataset must be first stored and archived in the form of a file. Three file formats are widely physics laboratory exports:

1. Comma-separated values (CSV files)
2. Instrument-specific data (DAT files)
3. Microsoft Excel spreadsheets (XLSX files)

1.1 ASCII files

ASCII, which is an acronym for *American Standard Code for Information Interchange*, is a character encoding standard. Originally, it was used for representing a particular set of 95 printable characters and 33 control characters, thereby giving a total of 128 code points, using 7 bits. Later, an 8-bit encoding extended the ASCII set to a total of 256 characters.

An ASCII file is one that contains data made up of ASCII characters, i.e. it is essentially raw text. Each byte in the file contains one character that conforms to the standard ASCII code.

Source codes of computer programs, batch files, macros and scripts are straight text and stored as ASCII files. HTML and XML files are also ASCII files. Text editors such as Notepad create ASCII files as their native file format.

CSV and DAT files are two of the most commonly used ASCII file formats used for storing scientific data for subsequent analysis.

1.1.1 CSV files

A comma-separated values (CSV) file is a plain text data format for storing tabular data where the fields (values) of a record are separated by a comma and each record is a line (i.e. newline separated). For example, the following are the contents of a CSV file:

```

1      id, name, email
2      1, John, john.doe@example.com
3      2, Jane, janey72@test.org

```

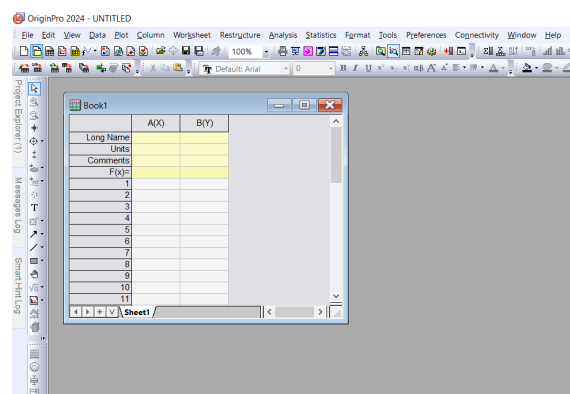
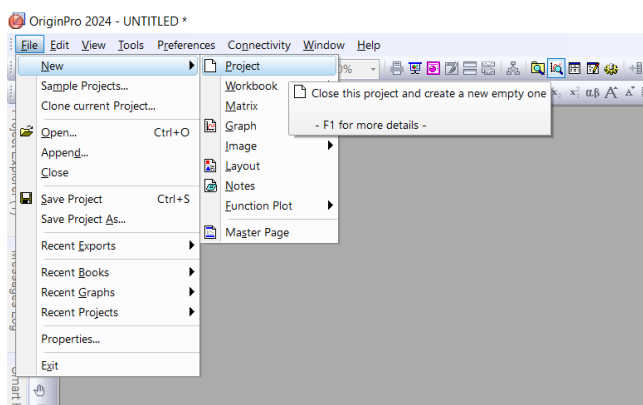
In software that generally deals with tabular data such as a database or a spreadsheet, the contents of the above CSV file is parsed in the form of a table as follows:

id	name	email
1	John	john.doe@example.com
2	Jane	janey72@test.org

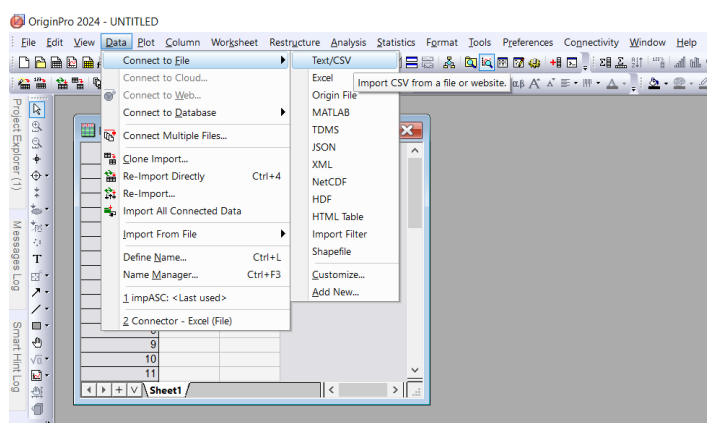
The first line is used as the table header and the remaining lines are used as the rows of the table.

Importing data from a CSV file in Origin:

- I. In the main menu of Origin¹, go to File > New > Project. This would create a new Origin project with an empty worksheet.

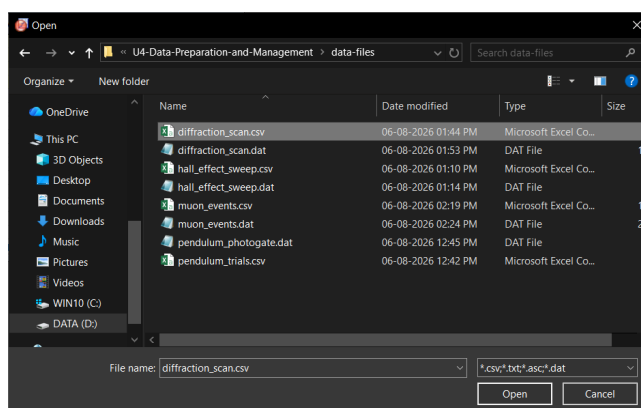


- II. To fill in the worksheet with values from the CSV file, we go to Data > Connect to file > Text/CSV.

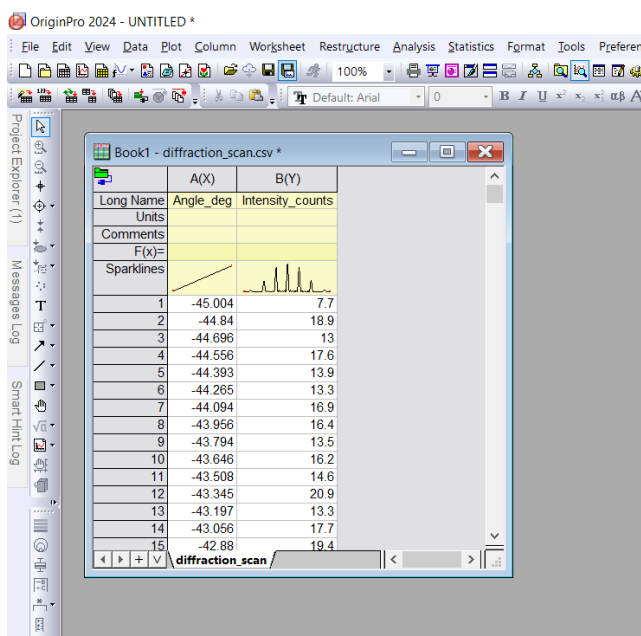
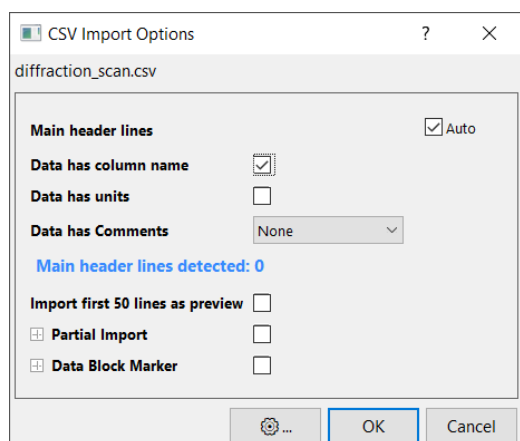


¹Documentation on the Origin graphical user interface (GUI)

III. In the file open dialog, we navigate to the folder containing the CSV file. Selecting the file of interest, we click Open.



IV. Another dialog appears where we can provide the options for importing data from the CSV file. Keeping everything unchanged, we click OK. We find that the data from the CSV file has been imported into the worksheet and is ready for further analysis.



1.1.2 DAT files

Computer programs create DAT files to store specific information or data. The information in the file is only relevant to the program that created it, though almost any program can create DAT files. DAT might look like an acronym, but the extension name is actually short for 'data.'

A DAT file may begin with header lines that begin with #. Any line that begins with a # is treated like a comment and is subsequently ignored. So we can make use of these header lines to insert specific information in the DAT file. However, it is possible to have DAT files without such header lines.

The actual data in a DAT file may be entered with each record in its own row and separated by whitespaces, such as space or tab. For example, the following are the contents of a DAT file:

```

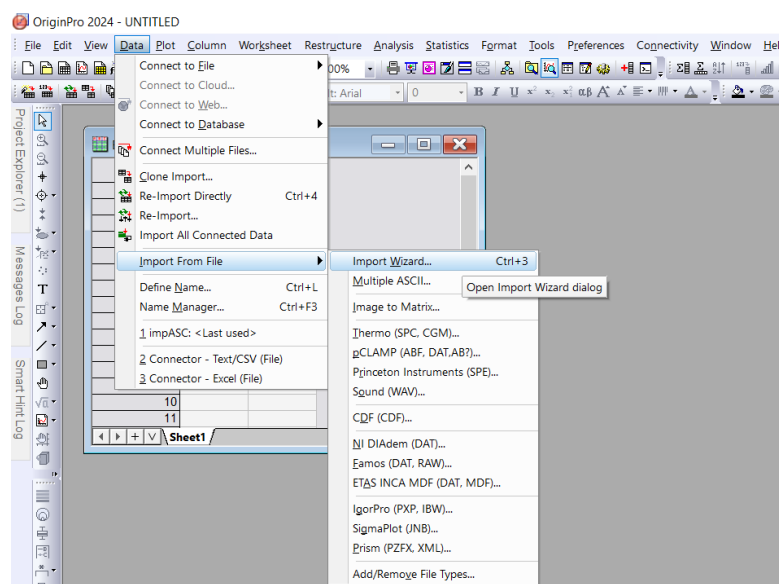
1      # Diffraction Grating Intensity Scan
2      # Wavelength_nm: 632.8
3      # Grating_spacing_nm: 3000.0
4      # Slit_width_nm: 800.0
5      # Date: 2026-08-01
6      Angle           Intensity
7      degree          counts
8      -45.000         10.5
9      -44.850          9.6
10     -44.700         16.3
11     -44.550         11.5
12     -44.400         15.2

```

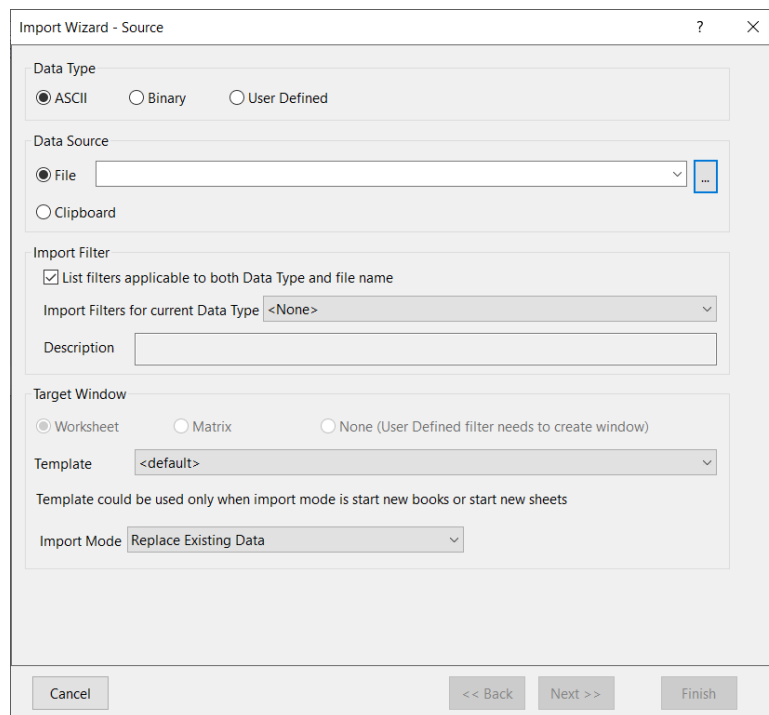
In the above DAT file, lines 1 to 5 are indicated as headers. Lines 6 and 7 are provide the column labels and units respectively. The subsequent lines contain the data, separated by a tab space.

Importing data from a DAT file in Origin:

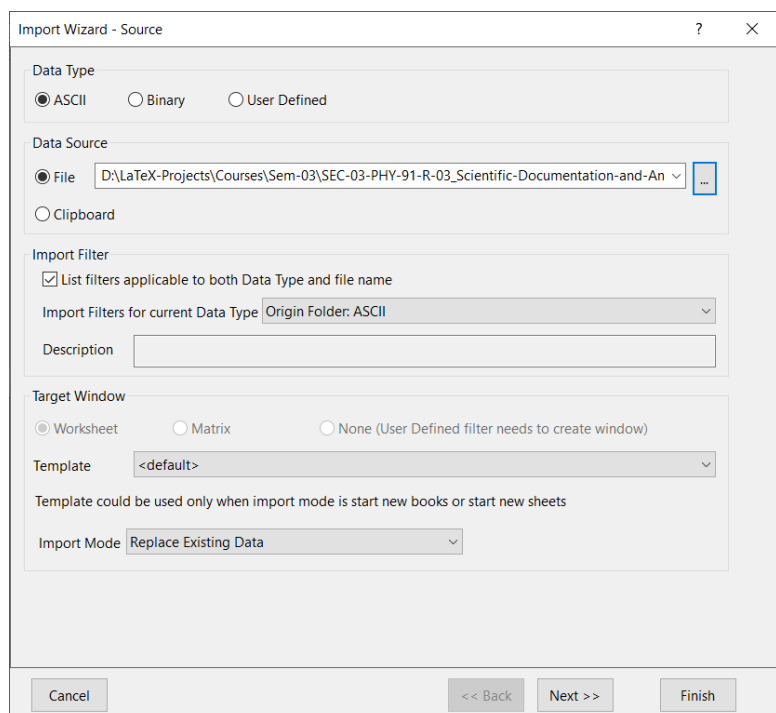
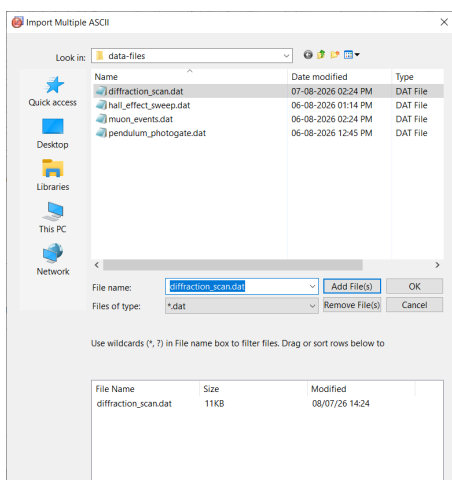
- I. Create a new project in Origin. Then to import data from a DAT file, go to Data> Import From File> Import Wizard.



II. In the Import Wizard dialog box, we ensure that under Data Source, the File option is checked and then we click on the button to the right of the filename textbox.



III. After navigating to the folder that contains the DAT file of interest, we select the file (or multiple files). Then we click Add File(s) and finally OK. We would be taken back to Import Wizard dialog where we can confirm that the full path of the selected DAT file is displayed in the filename textbox.



IV. Clicking Next on the Import Wizard dialog takes us to the next window where we set the options for data import.

We uncheck the Auto determine header lines checkbox. Then we enable the By Leading Character radio button and enter # in the textbox alongside. This is because in our DAT files, the header lines are indicated by the character # at the beginning.

We enter the input Number of subheader lines = 2 because two lines have special meanings as containing the long names of the columns and their units. We indicate these by the inputs Long Names = 1 and Units = 2.

A preview of the DAT file at the bottom of the dialog page confirms that our options are correct by displaying the header lines in black, the long names in blue, the units in green and the data in grey.

Import Wizard - Header Lines

☐ Auto determine header lines Main Header ☐ Fixed Number 5 ☒ By Leading Character #

Number of subheader lines 2 ☐ Line number start from bottom

Column Label Assignment from Subheader Lines

Long Names 1 Comments <None> to 0

Units 2 System Parameters <None> to 0

User Parameters <None> to 0

☐ Extract Long Names and Units from Same Line Characters to skip on each line 0

Preview Font System Preview Lines 50

Prefix: S=Short Name, L=Long Name, U=Units, P=Parameters, C=Comment, MH=Main Header, SH=Subheader

```

001 001MH      # Diffraction Grating Intensity Scan
002 002MH      # Wavelength_nm: 632.8
003 003MH      # Grating_spacing_nm: 3000.0
004 004MH      # Slit_width_nm: 800.0
005 005MH      # Date: 2026-08-01
006 001SH L    Angle
007 002SH U    degree
008           counts
009           10.5
010           9.6
011           16.3
012           11.5
013           15.2
014           12.2
015           11.4
016           16.3
017           8.1
  
```

Cancel << Back Next >> Finish

V. We click on Next for the remaining dialog pages and finally click on Finish.

Import Wizard - Save Filters

Import Wizard settings can be saved to a filter file for re-use. Filter files can be selected on the first page of this wizard. Once saved, import filters can also be used to automatically determine import settings when dragging and dropping data files into Origin and when opening data files with the File/Open menu item.

☒ Save filter

☐ In the data file folder D:\LaTeX-Projects\Courses\Sem-03\SEC-03-PHY-91-R-03_Scientific-Documenta

☒ In the User Files folder C:\Users\Asus\Documents\OriginLab\User Files\Filters\

☐ In the Window

☐ Show Filter in File/Open List

Filter Description

Filter file name (.OIF extension will be appended) diffraction_scan

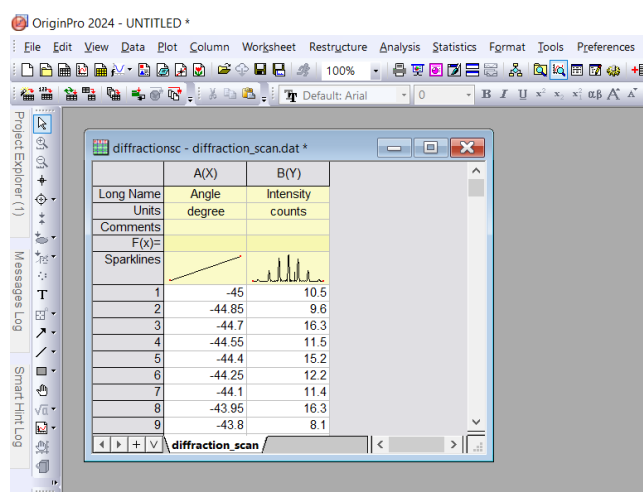
Applicable File diffraction_scan*.dat

Use wild cards such as *.txt, s???*.dat etc. Specify multiple names by ;, such as a*.dat; b*.dat

☐ Specify advanced filter options

Cancel << Back Next >> Finish

VI. When the Import Wizard closes, we find that the data from the DAT file has been correctly imported into the Origin workbook and it is now ready for analysis.



1.2 Interoperability using spreadsheets

1.2.1 XLSX files

An XLSX file is the native workbook format of Microsoft Excel. It is also read and written compatibly by most other spreadsheet platforms, such as LibreOffice Calc and Google Sheets.

Structurally, an XLSX file is fundamentally different from the CSV and DAT files because it is not an ASCII file. Rather than being a flat sequence of human-readable characters, an XLSX file is a compressed (ZIP) container holding several internal XML documents that together describe one or more sheets, along with their formatting, formulas, and metadata. Hence, it cannot be usefully opened in a plain text editor the way a CSV or DAT file can.

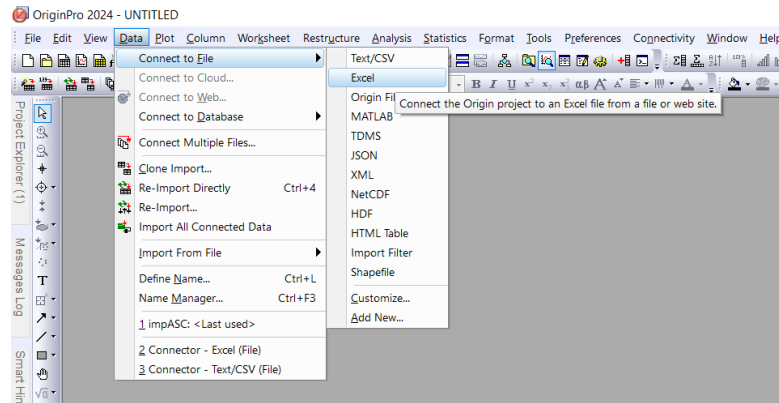
A CSV or DAT file can be parsed by any program capable of reading text, provided the delimiter and header conventions are known. An XLSX file, by contrast, can only be read by software that implements the Office Open XML specification (or a compatible library); Origin does this via a dedicated Excel import module rather than the ASCII import wizard.

A single XLSX workbook may contain multiple sheets, each an independent two-dimensional grid of cells. Unlike an ASCII file, a sheet can, by itself, carry more than one header row without any special import configuration. For example, a column-name row followed by a units row.

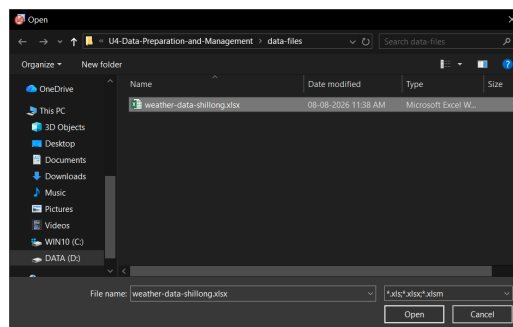
1.2.2 Working with XLSX files in Origin

Importing data from an XLSX file

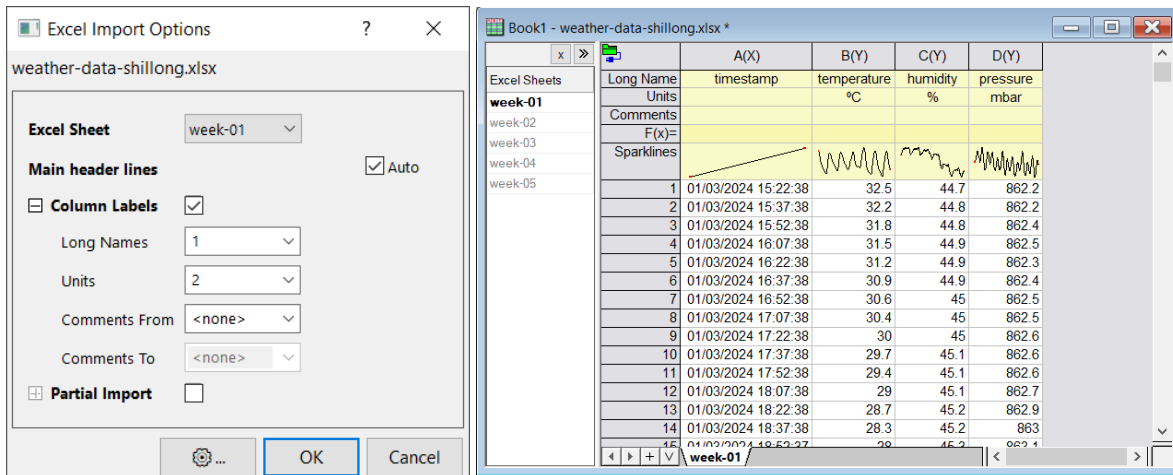
- I. Create a new project in Origin. Then to connect to an XLSX file, go to **Data > Connect to File > Excel**.



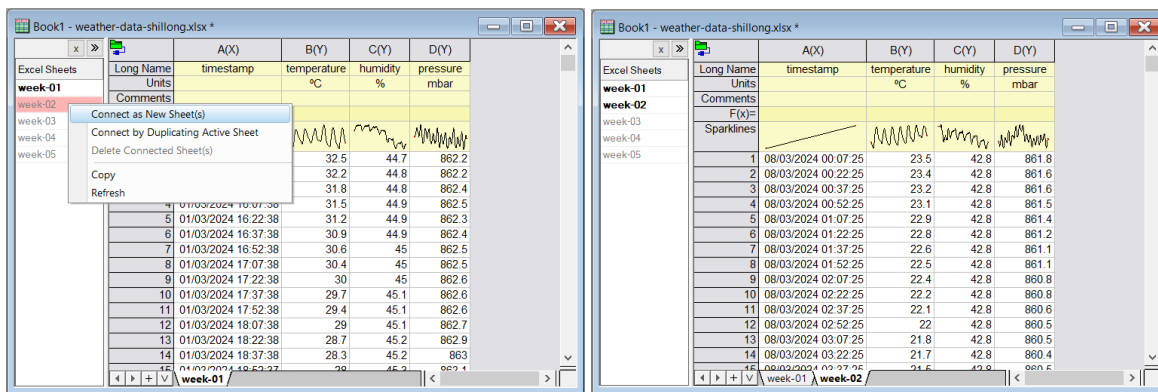
- II. After navigating to the folder that contains the XLSX file of interest, we select the file and then click Open.



- III. In the Excel Import Options dialog, we set Excel Sheet as any one of the sheets, usually the first. Also we set Long Names = 1 and Units = 2. Then on clicking OK, we find that the data from the selected sheet has been imported into the workbook. However, we also find that, in the Excel Sheets sidebar, the other sheets are listed, although they are greyed out, which means that their data has not been imported.



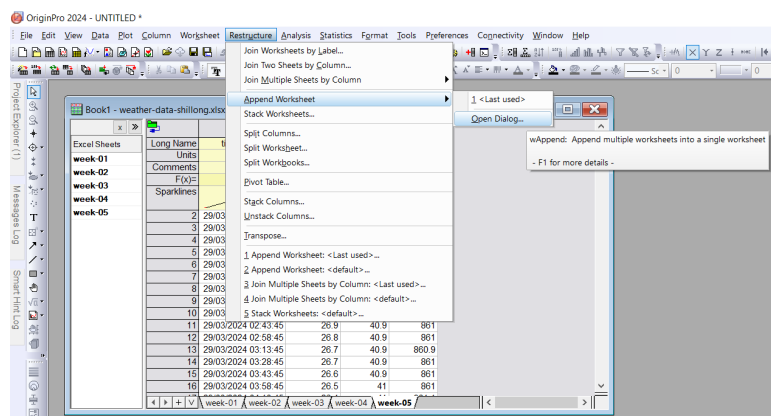
IV. To import data from another sheet of the XLSX file, we select that sheet from the sidebar, then right-click and select **Connect as New Sheet(s)**. This will import its data into a new worksheet.



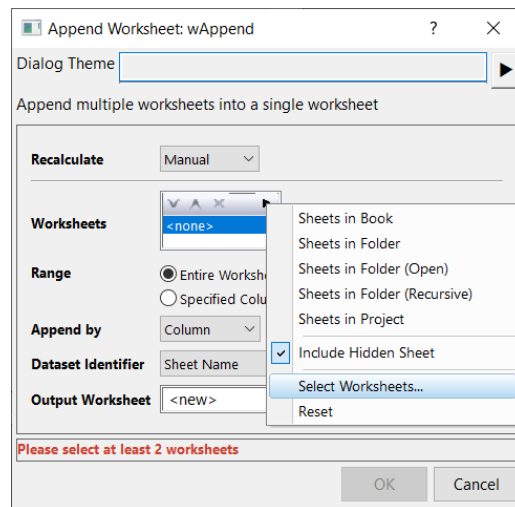
Likewise, we can import data from all the remaining sheets into separate worksheets.

Appending data from multiple sheets

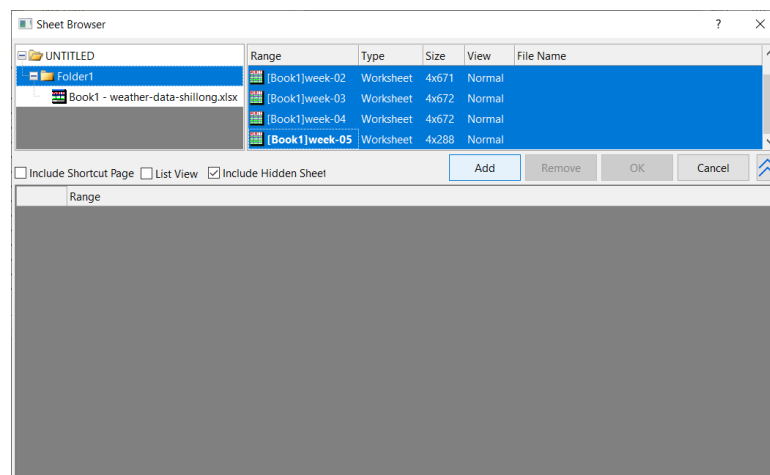
I. Suppose that we have similar data in the different sheets of an XLSX file that we have imported. Now to combine the data into a new sheet, we go to **Restructure > Append Worksheet > Open Dialog**.



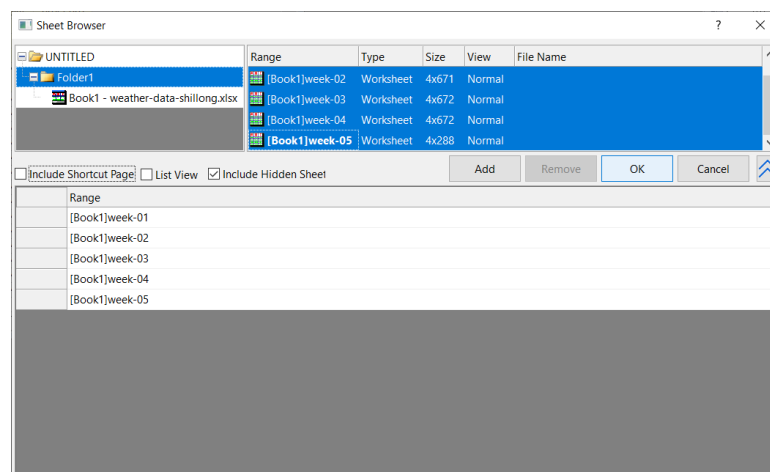
- II. On the Append Worksheet dialog, click on the triangle button to the right of the Worksheets option and then click Select Worksheets.



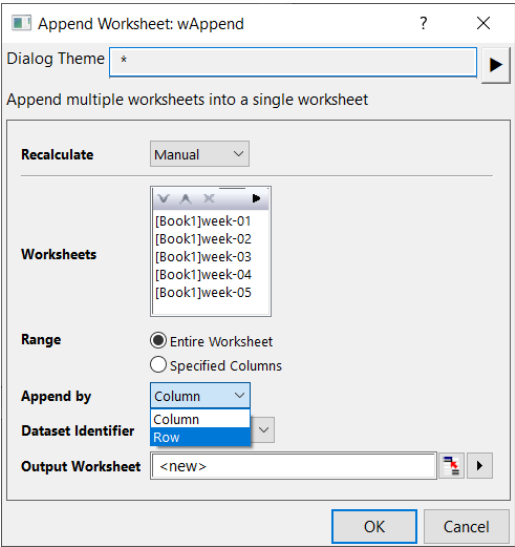
- III. The Sheet Browser dialog would then open up. Pressing and holding down the Ctrl button, click and select all the worksheets, and then click Add.



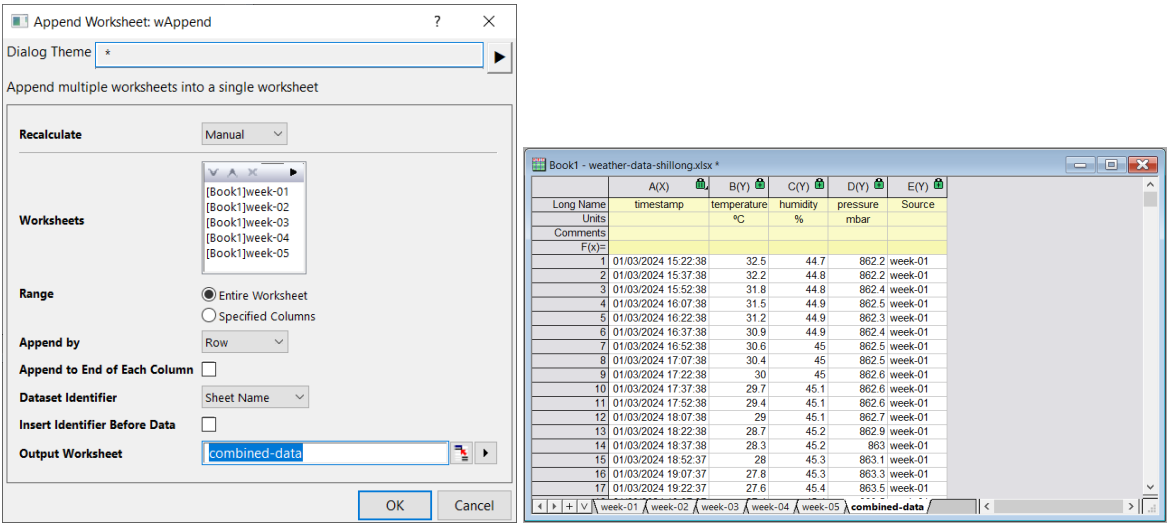
- IV. The selected worksheets would then appear in the list below. We then click OK.



V. After the above step, we would be back in the Append Worksheet dialog. We set Append by = Row.



VI. In Output Worksheet we type the name of the new worksheet which is to contain the combined data. Then we click OK to find that a new worksheet has been created, which contains the combined data from all the worksheets.



2 Masking and filtering of data

In physical laboratory experiments, data recorded by instruments often contains unwanted noise, electrical spikes, or out-of-range readings. Filtering and masking are two fundamental techniques used to clean data before plotting or mathematical fitting.

2.1 Masking data

In computational data analysis software like Origin, masking operates at the data cell or point level by attaching an internal metadata boolean flag (a mask status bit) to specific entries within a data structure. The raw numerical array remains entirely intact in memory. Masking does not alter, reindex, or delete row entries from the underlying data vector.

When statistical routines (such as mean, variance, or polynomial fitting) or graphic rendering engines query the dataset, they check the mask attribute of each cell. If a cell's mask flag is evaluated as TRUE (active mask), the processing engine skips that memory address during iteration or sets its inclusion weight to zero.

Masking allows arbitrary cell-level selection regardless of contiguous row order. On the user interface layer, the software highlights masked cells with an overlay color (typically red) and excludes them from graph series without breaking the continuity of the remaining row index mappings.

2.1.1 Masking points in Origin

- I. Open the worksheet containing the dataset.
- II. Highlight the specific cells or rows that are to be masked and do any of the following:
 - (a) Right-click the highlighted selection and click Mask > Apply.
 - (b) From the main menu, go to Column > Mask > Apply.
 - (c) On the shortcut menu, toggle the Mask/Unmask button.
 - (d) Click on the Mask Range button on the mask toolbar.
- III. The background of the selected cells turns red. The colour of the masked cells can be changed by clicking on the Change mask color button on the toolbar.
- IV. Any plot created from this dataset will indicate the masked cells in the same colour. The masked data points can also be hidden from the plot by toggling the Hide/Show masked points button on the toolbar.
- V. To remove the mask, highlight the masked cells and do any of the following:
 - (a) Right-click the highlighted selection and click Mask > Remove.
 - (b) From the main menu, go to Column > Mask > Remove.
 - (c) On the shortcut menu, toggle the Mask/Unmask button.
 - (d) Click on the Unmask Range button on the mask toolbar.

2.2 Filtering data

Filtering operates as a relational query or dynamic predicate evaluation applied across entire records (rows) of a dataset based on logical conditions defined for one or more variables.

The software evaluates a conditional logical expression, such as $f(x_i) = \text{TRUE}$ for $x_i \geq x_{\text{threshold}}$, across a specified column vector $\mathbf{x}$. This produces a global boolean mask vector corresponding to all rows in the worksheet.

Filtering creates a subset view (a temporary relational selection) of the worksheet. Rows where the condition evaluates to FALSE are collapsed or hidden in the user interface and bypassed by downstream analysis routines.

Unlike cell-level masking, filtering acts on full rows across all columns simultaneously to preserve tabular record integrity. When the filter rule is toggled off or modified, the full dataset view is restored instantaneously because no data points were modified or unlinked.

2.2.1 Filtering points in Origin

- I. Open the worksheet containing the dataset.
- II. Click on the column on which the filter is to be applied and do any of the following:
 - (a) From the main menu, go to `Column > Filter > Add or Remove Filter`.
 - (b) Click on the `Add/Remove data filter` button on the worksheet data toolbar.
- III. Once a filter has been applied to a column, click on the filter icon on the top-left of the column header and click on the appropriate option from the context menu to modify the filter.
- IV. A filter can be enabled or disabled by selecting the filtered column and clicking on the `Enable/Disable data filter` button on the worksheet data filter.
- V. To remove a filter, click on the filter icon on the column header and then click `Clear Filter`.

Exercises

1. Import `muon_events.csv` into a fresh worksheet. Apply a data filter on `Pulse_Height_mV` to display only high-energy events where the pulse height is strictly greater than 50.0 mV. How many total rows remain visible after applying this filter?
2. Import `diffraction_scan.csv`. In optical diffraction measurements, detector saturation occurs at an intensity of 4095.0 counts. Identify the saturated rows around 0° angle where `Intensity_counts` equals 4095.0 and manually mask them.

3 Sorting of data

Sorting means arranging the rows of a dataset in a specific order—either ascending (*smallest to largest* or *A to Z*) or descending (*largest to smallest* or *Z to A*) based on the values in a selected column.

Instruments do not always log data in strict chronological or numerical order. Sorting helps organize observations by an independent variable (such as magnetic field strength, angle, or time) so that line plots connect data points continuously rather than crisscrossing back and forth across the graph.

3.1 Sorting worksheet rows in Origin

- I. Open the worksheet containing the dataset.
- II. Click on the column using which the rows of the worksheet need to be sorted and do any of the following:
 - (a) From the main menu, go to `Worksheet > Sort Worksheet > Ascending or Descending` or any other customised sorting criteria with the `Custom` option.
 - (b) Right-click on the column header and then go to `Worksheet > Sort Worksheet > Ascending or Descending` or any other customised sorting criteria with the `Custom` option.

Exercises

1. Import `hall_effect_sweep.csv`. Sort the entire worksheet in descending order based on the `Field_mT` data. What is the maximum field strength value?
2. Import `muon_events.csv`. Sort the worksheet in ascending order by the data in the `Pulse_Height_mV` column. What timestamp corresponds to the lowest pulse height?

4 Normalization of data

Normalization is the process of scaling numeric values measured in arbitrary units into a standardized relative range, typically between 0 and 1 (or 0% and 100%).

In optical spectroscopy, X-ray diffraction (XRD), or gamma-ray spectra, raw peak heights depend on lamp power, detector gain, or exposure time. To compare the shapes or relative peak ratios of two different experimental scans, we divide all data values by the maximum peak value:

$$Y_{\text{normalized}} = \frac{Y}{Y_{\text{max}}}$$

When background baseline offset is present, we adjust the scale so the minimum value becomes 0 and the maximum becomes 1. This is known as *min-max scaling* as follows:

$$Y_{\text{normalized}} = \frac{Y - Y_{\text{min}}}{Y_{\text{max}} - Y_{\text{min}}}$$

4.1 Normalizing data in Origin

4.1.1 First method (in-built method)

- I. Open the worksheet containing the dataset.
- II. Click to highlight the entire column that contains data to be normalized, and do any of the following:
 - (a) From the main menu, go to Analysis > Mathematics > Normalize Columns > Open Dialog.
 - (b) Right-click on the column header and, on the context menu, click on Normalize.
- III. Select the appropriate option against Normalize Methods.
The default is Normalize to [0, 1]. Then click OK.
- IV. The normalized data is calculated and placed in a new column. The long name of the column can then be edited as per requirement.

4.1.2 Second method (formula method)

- I. Open the worksheet containing the dataset.
- II. Right-click on the empty area to the right of the right-most column, and then click Add New Column.
- III. A new column with created, say it is named C. It can be given an appropriate long name.
- IV. Click to highlight the new column C. Then right-click and select Set Column Values.
- V. The Set Values dialog opens up. Suppose we want to fill up column C with normalized values from column B, then in the formula box, we type the following formula and click OK:

$$(Col(B) - Min(Col(B))) / (Max(Col(B)) - Min(Col(B)))$$

Exercises

1. Using `diffraction_scan.csv`, perform normalization on Column B (i.e. `Intensity_counts`) using the Origin's in-built method.
2. Import `muon_events.csv`. Create a new column titled `Normalized_Pulse` and scale `Pulse_Height_mV` using the formula method.